Multilingual Continuous User Authentication Using Keystroke Dynamics

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Abstract

Continuous user authentication based on behavioural biometrics has received increasing attention as a supplement to traditional password-based security systems. This study examines the effectiveness of keystroke dynamics for multilingual user authentication, with particular emphasis on the impact of language variation on typing behaviour and authentication accuracy. A machine learning evaluation framework has been developed to analyse a publicly available multilingual keystroke dynamics dataset, incorporating feature extraction, data preprocessing, and comparative model assessment. Multiple classification algorithms have been trained and evaluated using biometric-specific performance metrics. Currently, a modular evaluation pipeline has been established, baseline models have been trained, and initial cross-language analyses have been performed, providing a foundation for the final experimental evaluation.

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Introduction

Background

As digital systems increasingly underpin critical aspects of everyday life, secure and reliable user authentication has become a fundamental requirement. Traditional authentication mechanisms, such as passwords and PINs, remain the most widely deployed methods despite well-documented weaknesses. These include poor memorability, password reuse across platforms, vulnerability to phishing and credential stuffing attacks, and limited effectiveness against automated threats. As a result, reliance on knowledge-based credentials alone is no longer sufficient to protect sensitive systems such as online banking platforms, educational environments, shared computing systems, and personal digital accounts.

Biometric authentication has emerged as a promising alternative, offering the ability to identify individuals based on inherent physiological or behavioural characteristics rather than memorised secrets. Physiological biometrics, including fingerprints, facial recognition, and iris scanning, are already deployed in many commercial and governmental systems. While these modalities provide strong identity assurance, they raise concerns related to privacy, permanence, and spoofing. Once compromised, physiological traits cannot be revoked or replaced, and advances in deepfake technology and high-resolution replication have further eroded confidence in their long-term security.

In contrast, behavioural biometrics focus on how users interact with systems rather than what physical traits they possess. Behavioural signals such as typing rhythm, mouse movement, handwriting dynamics, and gait patterns are continuously produced during normal interaction and are therefore well suited to transparent and continuous authentication. Among these modalities, keystroke dynamics has attracted significant research attention due to its non-intrusive nature, low deployment cost, and compatibility with existing hardware. By analysing timing information generated during typing, keystroke dynamics enables identity verification without requiring additional sensors or explicit user action.

Recent research has demonstrated that keystroke dynamics can achieve high authentication accuracy across both fixed-text and free-text scenarios, and can be applied in domains ranging from cybersecurity and banking to education and academic integrity. However, despite these advances, existing systems often operate under restrictive assumptions, such as monolingual usage, stable typing conditions, and static user behaviour. These assumptions limit the applicability of keystroke-based authentication in real-world contexts characterised by multilingual users, variable cognitive load, and evolving interaction patterns.

First, much of the existing research is biased toward English-language datasets and does not adequately address multilingual typing behaviour. In practice, many users routinely type in multiple languages, switch between scripts, or combine languages within a single task. These behaviours introduce variability in key usage, rhythm, and cognitive processing that can degrade authentication performance if not explicitly accounted for.

Second, typing behaviour is influenced not only by motor skills but also by cognitive load and task complexity. Writing tasks that involve planning, evaluation, or creative synthesis produce different keystroke patterns from tasks involving simple transcription or recall. Most authentication systems treat deviations from a baseline as potential security threats, without distinguishing between legitimate behavioural variation and impostor activity. This increases the risk of false rejections and reduces system usability.

Third, many high-performing keystroke authentication models prioritise classification accuracy without sufficient consideration of adaptability, transparency, or ethical implications. Black-box deep learning approaches, while powerful, often require large training datasets, lack interpretability, and may struggle to adapt to gradual behavioural drift without retraining. In security-sensitive environments, such limitations can undermine trust and hinder adoption.

Finally, modern digital ecosystems increasingly demand continuous authentication rather than single-point verification. Users remain logged into systems for extended periods, during which credentials may be compromised or devices shared. Without continuous behavioural monitoring, traditional authentication models provide limited protection once initial access is granted.

These challenges highlight the need for an adaptive, multilingual, and continuous keystroke dynamics framework that balances security, usability, and inclusivity.

Aims and objectives

The objective of this project is to investigate and design a behavioural biometric authentication framework based on keystroke dynamics that supports continuous authentication in realistic, multilingual usage scenarios. The project pursues the following objectives to achieve the main objective:

- Critically examine current biometric authentication methods, focusing especially on behavioural biometrics and keystroke dynamics.
- Analyse and compare modelling techniques for keystroke-based authentication, including statistical, machine learning, and hybrid methods.
- Investigate how multilingual typing and cognitive load affect keystroke dynamics.
- Identify the limitations of current systems in terms of adaptability, fairness, and real-world use.
- Propose a system design that enables continuous authentication and reduces false rejections from normal behavioural changes.
- Define both functional and non-functional requirements to guide the development of a strong and ethical authentication solution.

Approach

This project uses an iterative and flexible development method based on agile principles. Agile methods suit the exploratory nature of behavioural biometric research, where system design must be refined as new findings emerge. Instead of following a strict, step-by-step process, the project focuses on gradual analysis, evaluation, and improvement.
The research part of the project is based on a thorough review of literature about biometric authentication, keystroke dynamics, behavioural modelling, multilingual interaction, and continuous authentication systems. This background guides system design choices and keeps the project in line with current academic and industry standards.
Technically, the project looks at keystroke timing features instead of the actual text, which helps protect user privacy and makes the system language-independent. The modelling methods aim to be both strong and easy to understand, so the system can adjust to real changes in user behaviour while staying secure.
Risk assessment and ways to reduce risks are included at every stage of the project, covering issues like limited data, changes in behaviour, and complex integration. Ethical issues, such as using minimal data and being transparent, are treated as key parts of the design, not as afterthoughts.

Literature Review

2.1 Approach to literature searching

Authentication mechanisms have traditionally relied on knowledge-based factors such as passwords and PINs. However, extensive research has demonstrated that these mechanisms suffer from fundamental usability and security limitations, including weak password selection, reuse across platforms, and susceptibility to phishing and credential stuffing attacks (Human biometric identification: Application and evaluation, 2024). As digital services increasingly handle sensitive personal, financial, and academic data, the inadequacy of password-only authentication has become a critical concern. Biometric authentication

addresses these limitations by identifying users based on inherent physiological or behavioural traits. Physiological biometrics—such as fingerprints, facial recognition, iris, and retinal scans—are widely deployed due to their high uniqueness and accuracy (Human biometric identification: Application and evaluation, 2024). However, these modalities raise significant privacy and security concerns. Physiological traits are permanent, cannot be revoked if compromised, and have been shown to be vulnerable to spoofing through high-resolution replication and deepfake technologies (Bora et al., 2025).

Behavioural biometrics offer an alternative approach by focusing on how users interact with systems rather than what physical characteristics they possess. Traits such as gait, mouse movement, handwriting dynamics, and keystroke dynamics are continuously produced during normal interaction, enabling transparent and ongoing authentication (BabaAli, 2021). Among these, keystroke dynamics has emerged as one of the most practical and cost-effective behavioural modalities due to its compatibility with standard keyboards and minimal hardware requirements.

2.2 Identifying the problem

Early research on keystroke dynamics focused on whether typing behaviour could serve as a reliable behavioural biometric outside of controlled, single-language settings. Foundational studies showed that keystroke dynamics can achieve high authentication accuracy in fixed-text and single-language situations (Monrose and Rubin, 2000; Killourhy and Maxion, 2009). However, more recent literature highlights significant limitations when these systems are used in multilingual, real-world environments.

One main issue is that most research assumes users type in the same language during both enrolment and authentication. Most keystroke dynamics studies use data collected in a single language, usually English, and test models under the same conditions (Banerjee et al., 2023). Yet, evidence shows that typing behaviour depends not just on motor skills but also on language structure, writing systems, and cognitive demands. Studies on cross-language input find noticeable changes in timing and more errors when users switch languages or scripts, which reduces authentication accuracy (Giot et al., 2022; BabaAli, 2021). This raises questions about using monolingual keystroke models for multilingual users.

Further analysis shows that researchers have not closely examined how stable keystroke features are across different languages. Some recent methods use advanced deep learning to boost accuracy, but these models usually treat language differences as background noise instead of something to study directly (TempCharBERT, 2023). As a result, it is still unclear which keystroke features, like dwell time or inter-key latency, stay the same across languages and which ones change. This lack of systematic cross-language comparison is a major gap in current research.

The review also found that experimental design and reporting are often inconsistent. Many studies report only general classification accuracy, while important biometric metrics like Equal Error Rate (EER), False Acceptance Rate (FAR), and False Rejection Rate (FRR) are rarely or inconsistently used (Jain et al., 2024). Without standard evaluation methods, it is hard to compare systems or judge if they are suitable for security-focused authentication.

Taken together, these findings point to a clear research problem. While there is strong evidence that keystroke dynamics can work as a behavioural biometric, we still do not fully understand how reliable or generalisable it is in multilingual settings. To address this, we need structured, metric-based evaluations of multilingual keystroke datasets using consistent machine learning methods.

2.3 Keystroke Dynamics as a Behavioural Biometric

Biometric authentication systems work by identifying or verifying people using unique physical or behavioural traits. Traditional methods like fingerprints, iris scans, and facial recognition are accurate, but they often need special equipment and can raise privacy issues (Sajjad et al., 2022; Roy et al., 2022). Behavioural biometrics, on the other hand, look at how users interact with systems, providing a non-intrusive and continuous form of authentication. Among these, keystroke dynamics has emerged as a promising

behavioural biometric due to its low cost, transparency, and compatibility with existing devices (Roy et al., 2022).

Keystroke dynamics is the study of how people type, focusing on timing features like how long a key is held down (dwell time), the time between keystrokes (flight time), and the intervals between pairs or groups of keys (Altwaijry, 2023). These patterns reflect unconscious motor habits that are hard to fake, which helps protect against impersonation attacks (Martins et al., 2025). Unlike static methods, keystroke dynamics can be used for both single and ongoing checks, so systems can confirm identity throughout a session, not just at login.

Early keystroke dynamics systems mostly used fixed text, like passwords or set phrases, and worked well in controlled settings (Roy et al., 2022). But these methods do not work as well in real situations where people type freely. Because of this, recent studies have shifted toward free-text and continuous authentication models that better match how people actually use computers (Arsha et al., 2024).

2.3.1 Machine Learning Approaches in Keystroke Authentication

The effectiveness of keystroke dynamics authentication depends on the machine learning techniques employed to model typing behaviour. Classical statistical models, including Gaussian Mixture Models (GMM) and Hidden Markov Models (HMM), are widely adopted for their capacity to address temporal variability and limited training data (Roy et al., 2022). Notably, GMM-based systems exhibit strong performance in continuous authentication scenarios, particularly when integrated with probabilistic distance measures such as the Bhattacharyya distance (Sajjad et al., 2022).

Recent research has investigated discriminative machine learning techniques, such as Support Vector Machines (SVM), Random Forests, and ensemble classifiers. Martins et al. (2025) found that SVM-based classifiers achieve balanced False Acceptance Rate (FAR) and False Rejection Rate (FRR) values in biometric authentication tasks, indicating their suitability for security-sensitive applications. Ensemble learning methods further enhance robustness by integrating multiple classifiers to reduce model bias and variance (Arsha et al., 2024).

Deep learning models have become increasingly prominent, especially for large-scale and free-text datasets. Roy et al. (2022) reported that deep neural networks and ensemble DNN architectures significantly outperform traditional models when ample training data is available. Additionally, TempCharBERT-based architectures show that integrating temporal keystroke features into language models substantially improves authentication accuracy by capturing both linguistic and behavioural patterns (BabaAli, 2021). However, deep learning models present challenges regarding data requirements, computational complexity, and interpretability.

2.3.2 Language Dependency in Keystroke Dynamics

While keystroke dynamics is often considered language-independent, research shows that typing behaviour is influenced by the language used. Typing in different languages changes cognitive load, keyboard usage, character frequency, and motor planning, all affecting keystroke timing features (Altwaijry, 2023). This challenges the assumption that a single keystroke profile can reliably authenticate a user across multiple languages.

Altwaijry (2023) conducted a comprehensive study on language influence in keystroke dynamics, showing measurable performance degradation when users were authenticated in a language different from their enrolment language. Equal Error Rates increased notably in cross-language scenarios, especially when switching between languages with distinct scripts or typing conventions. These findings highlight a critical limitation in many existing keystroke authentication systems, which assume linguistic consistency.

Multilingual and bilingual datasets show that users have distinct typing rhythms when switching languages, even on the same keyboard (Tsimperidis et al., 2024). Differences in character distributions, digraph frequencies, and typing familiarity contribute to these variations. Studies on

non-Latin scripts, such as Arabic and Chinese, show even greater divergence in keystroke patterns due to different keyboard layouts and input methods (Ciślik et al., 2024).

2.3.4 Cross-Language and Multilingual Authentucation Challenges

Multilingual authentication must balance security, usability, and adaptability. Traditional methods often create separate keystroke profiles for each language, increasing enrolment burden and reducing user experience (Altwaijry, 2023). Keeping multiple profiles also brings security risks, since old templates can lead to more false acceptances or rejections.

Recent research has explored alternative strategies for multilingual differences. Domain adaptation and transfer learning try to find features in user behavior that stay the same across languages (Martins et al., 2025). Adaptive template updating has also been suggested, so models can slowly adjust to new language patterns and lower the risk of impostor adaptation (Roy et al., 2022).

Large studies on continuous authentication show that timing features, like dwell and flight times, handle language changes better than features based on key location or content (Tsimperidis et al., 2024). This suggests that multilingual keystroke authentication systems should focus on behavioral features that do not depend on language.

2.4 Continuous Authentication and Ethical Considerations

Despite its advantages, keystroke dynamics raises ethical and privacy concerns due to the potential for unintended data inference. Timing data may reveal sensitive information, including typed content or cognitive states (Sajjad et al., 2022). To mitigate these risks, numerous studies advocate for anonymisation techniques that discard keycodes while retaining timing features, thereby enabling behavioural analysis without content reconstruction (Roy et al., 2022).

Keystroke dynamics provides significant advantages over hardware-based biometrics, particularly in remote and online environments such as education, banking, and virtual examinations. Continuous authentication frameworks have demonstrated effectiveness in detecting impersonation during online assessments without disrupting legitimate users (Sajjad et al., 2022). These applications drive research into multilingual models, as global systems increasingly serve diverse populations.

Conclusions

The reviewed literature establishes keystroke dynamics as a mature and effective behavioural biometric for continuous user authentication. Although machine learning and deep learning have enhanced authentication accuracy, existing systems frequently overlook the impact of language on typing behaviour. Empirical evidence demonstrates that language switching introduces measurable variability in keystroke patterns, resulting in degraded authentication performance in cross-language scenarios.

Although several studies acknowledge this limitation, current solutions frequently depend on language-specific profiles or controlled experimental settings, which limits their practicality in real-world multilingual environments. A clear research gap remains in developing authentication systems capable of reliably authenticating users across multiple languages without excessive enrolment overhead or security compromise.

This project directly addresses this gap by investigating multilingual continuous authentication using keystroke dynamics. By evaluating cross-language performance and exploring adaptive modelling strategies, this work builds upon existing research and contributes a focused, practical solution aligned with contemporary cybersecurity requirements.

Product Research

In this project, the "product" is defined as a **keystroke dynamics authentication program** designed to evaluate multilingual user identification performance using machine learning techniques. Rather than developing a commercial application, the project focuses on the **design, implementation, and systematic evaluation of authentication algorithms** under multilingual typing conditions.The system functions as an experimental testbed that captures keystroke timing data, extracts behavioural biometric features, trains machine learning models, and evaluates their effectiveness when users type in different languages. This approach is consistent with contemporary academic research in cybersecurity and behavioural biometrics, where the primary contribution lies in **algorithmic performance analysis rather than end-user software deployment** (Roy et al., 2022).

3.1 Existing Keystroke Dynamics Evaluation Frameworks

Most keystroke dynamics systems are tested in controlled experiments instead of real-world products. These experiments usually include data collection, feature extraction, classification, and performance evaluation.
Earlier research shows these evaluation programs can accurately measure authentication with both fixed and free text (Martins et al., 2025). Still, most systems are tested only in one language, which makes them less useful for multilingual settings. Cross-language testing is usually a side experiment, not a main goal (Altwaijry, 2023).
There are few evaluation frameworks designed for multiple languages. This project aims to fill that gap by creating a system that supports language-aware and cross-language testing.

3.2 Feature Extraction for Multilingual Keystroke Analysis

Keystroke dynamics authentication systems work best when the right timing-based features are chosen. Most evaluation frameworks focus on extracting these features:

- **Dwell Time**: Duration of key press
- **Flight Time**: Time between consecutive key events
- **Digraph and Trigraph Latencies**: Timing patterns across character sequences
- **Total Typing Duration**

While these features are inherently language-independent, they can be indirectly influenced by linguistic structure, character frequency, and cognitive load (Tsimperidis et al., 2024). Therefore, multilingual evaluations should consider potential shifts in feature distributions when users change languages.
Research shows that timing-based features are partially stable across languages but display measurable shifts that affect classification performance (Altwaijry, 2023). This study evaluates same-language and cross-language authentication scenarios to identify which features generalise most effectively.

3.3 Machine Learning Algorithms in Existing Systems

Current keystroke dynamics evaluation programs use several machine learning algorithms, such as Support Vector Machines (SVM), Random Forests, Decision Trees, AdaBoost, and deep learning models such as neural networks (Roy et al., 2022; Martins et al., 2025).
Traditional classifiers like SVMs and Random Forests are often chosen because they work well with small datasets and are easy to interpret. They perform well in monolingual settings but tend to have higher error rates when used with multiple languages. Newer systems use deep learning to improve generalisation, but these models need much larger datasets and more computing power.
This project takes a comparative approach by analysing how different machine learning algorithms perform in multilingual settings, instead of recommending one model as the best for all cases. This method follows best practices in biometric system evaluation and allows for an objective analysis of performance.

3.4 Evaluation Methodologies in Related Systems

Researchers usually evaluate authentication programs in keystroke dynamics by using biometric performance metrics such as:

- Accuracy
- False Acceptance Rate (FAR)
- False Rejection Rate (FRR)
- Equal Error Rate (EER)

Of these metrics, EER is considered the most reliable for measuring authentication performance because it shows the balance between usability and security. Multilingual studies often find higher EER values in cross-language tests, which shows that switching languages is still a major challenge (Altwaijry, 2023).
Some evaluation frameworks use statistical analysis to compare how well systems perform in same-language and cross-language situations. This project uses a similar approach, with controlled experiments to measure how language changes affect authentication accuracy.

The review of existing evaluation programs shows that while keystroke dynamics authentication is a mature research area, multilingual evaluation is still insufficiently explored. Most frameworks lack systematic analysis of language switching and do not examine algorithmic adaptation across languages.
This project contributes by developing and evaluating a multilingual keystroke dynamics authentication program that explicitly measures cross-language performance. By analysing feature stability and machine learning behaviour across languages, the project aims to show how authentication systems can be more robust in multilingual environments.
The findings from this evaluation will directly inform the system design and requirements analysis, ensuring that the final implementation addresses real-world multilingual security challenges.

Legal, Social, Ethical and Professional Issues and Considerations

This project evaluates multilingual user authentication using keystroke dynamics by analyzing a publicly available dataset from GitHub. Since no primary data is collected, legal and ethical considerations focus on responsible secondary data use, compliance with data protection laws, and the broader impact of behavioural biometric technologies.

Legally, although the dataset is publicly accessible, keystroke dynamics data is a form of behavioural biometric information and may be considered personal data under the General Data Protection Regulation (GDPR) and the UK Data Protection Act 2018. Public access does not remove the obligation to process this data lawfully and responsibly. The dataset will be used only according to its published licence and intended research purpose. If documentation confirms anonymisation or informed consent, this will be acknowledged, and no attempts will be made to re-identify individuals or link behavioural patterns to real-world identities. The project follows the principle of purpose limitation, using the dataset only for academic evaluation of authentication algorithms, not for surveillance, profiling, or behavioural inference beyond identity verification. No additional data will be combined, and no attempts will be made to infer sensitive attributes such as emotional state, health, or personal habits. All analysis will be conducted locally, and only aggregated or statistical results will be reported to prevent reconstruction of individual contributors.
Social considerations relate to the impact of biometric authentication systems on diverse user populations. Keystroke dynamics systems may unintentionally disadvantage multilingual users if models are based mainly on dominant languages. This project addresses that risk by focusing on multilingual authentication performance, contributing to research that reduces bias and improves inclusivity. By evaluating algorithm performance across languages, the project supports the development of fairer and more representative authentication mechanisms.
Ethical concerns with behavioural biometrics include consent, transparency, and potential misuse, even when using open data. The project treats the dataset as sensitive, regardless of its public status. No raw data will be shared, and all visualisations or examples will be abstracted to prevent traceability. The project also critically evaluates the limitations and risks of keystroke dynamics, including false rejections and possible exclusion of legitimate users, rather than presenting it as a flawless security solution.
Professional standards guide the project's methodology and reporting. Following computing standards from the British Computer Society (BCS) and ACM, the project commits to transparency, reproducibility, and academic integrity. All datasets, tools, and algorithms will be properly cited, and any limitations or assumptions will be clearly acknowledged. The evaluation will avoid overstating security claims and will place findings within the context of existing biometric authentication research.
In conclusion, although this project does not involve direct interaction with human participants, the use of an open-source keystroke dynamics dataset still demands careful legal, ethical, social, and professional consideration. By respecting data licensing terms, avoiding re-identification, addressing bias in multilingual authentication, and adhering to established professional standards, the project ensures responsible and ethical use of behavioural biometric data while contributing meaningful insights to cybersecurity research.

Requirements

4.1 Analysis of Requirements

This project aims to evaluate how well keystroke dynamics–based authentication algorithms work in multilingual settings using machine learning. Unlike projects that focus on deploying systems or building user applications, this one is focused on analysis. The requirements cover data handling, algorithm evaluation, performance measurement, and meeting ethical and legal standards.
The system being tested is not meant for live authentication or commercial use. Instead, it serves as a controlled experiment that uses an existing dataset, extracts key features, applies machine learning, and

compares authentication results across languages. The main priorities are reproducibility, fairness, interpretability, and compliance, not real-time performance or user interface design

4.2 Comparison of systems

Keystroke dynamics authentication appears in many systems, each with its own goals and limits. Traditional systems often use **monolingual fixed-text authentication**, relying on password inputs and classic machine learning models such as Support Vector Machines or k-Nearest Neighbours. These systems usually assume users type in just one language and overlook the cognitive or motor differences involved in multilingual typing.
Recent systems now use keystroke dynamics for **free-text authentication**, continuous authentication, and deep learning-based identification. These systems are more flexible and accurate, but they often require large amounts of training data and are usually tested in only one main language, often English. As a result, it is still uncertain how well they perform with different languages.
Only a few recent studies have examined cross-language or multilingual typing behaviour. These studies show that typing rhythm depends not only on hardware, but also on language structure and mental effort. However, these systems are usually tested separately and lack a standard way to compare results. Specifically, the system under evaluation must:

- Support multilingual datasets rather than a single language
- Allow fair comparison between algorithms under identical conditions
- Avoid assumptions that typing behaviour is language-independent

4.3 Requirements elicitation

This project does not use primary requirements elicitation methods like questionnaires, interviews, or user studies. This choice is intentional because the work is analytical and uses an existing, publicly available dataset.
Instead, the requirements were gathered using the following approaches:

- Careful review of peer-reviewed literature on keystroke dynamics and multilingual authentication
- Review of the dataset's documentation and its limitations
- Study of ethical, legal, and professional standards for using biometric data
- Identification of gaps and limitations in current research on authentication

By using these secondary methods, the system requirements are based on established research and avoid the ethical risks of collecting behavioural biometric data.

4.4 Functional Requirements

The functional requirements define what the evaluation framework must be capable of doing in order to fulfil the objectives of the project.

1)Dataset Ingestion

The system shall be able to load and process a publicly available keystroke dynamics dataset obtained from GitHub.

2)Multilingual Data Handling

The system shall support datasets containing typing data from multiple languages and shall preserve language labels for comparative analysis.

3)Feature Extraction

The system shall extract standard keystroke dynamics features, including but not limited to:

- Key hold time (dwell time)
- Key flight time (up-down, down-down intervals)
- Timing-based rhythm features

4)Algorithm Implementation

The system shall implement multiple machine learning algorithms suitable for authentication evaluation, such as classical models (e.g., SVM, Random Forest) and/or neural network–based approaches where appropriate.

5)Training and Testing Pipeline

The system shall provide a repeatable training and testing pipeline that ensures consistent evaluation across languages and algorithms.

6)Performance Evaluation

The system shall compute standard biometric performance metrics, including accuracy, false acceptance rate (FAR), false rejection rate (FRR), and equal error rate (EER).

7)Cross-Language Comparison

The system shall enable comparison of authentication performance between different language subsets to assess multilingual robustness.

8)Result Reporting

The system shall output results in a structured format suitable for analysis, visualisation, and inclusion in an academic report.

4.5 Non-functional Requirements

Non-functional requirements describe the project's quality standards, constraints, and compliance needs.

1)Legal and Ethical Compliance

The system shall comply with GDPR principles and data protection regulations by using only anonymised, consented, and publicly licensed datasets, as outlined in the Legal, Social, Ethical and Professional Issues section.

2)Reproducibility

The evaluation process should be easy to repeat, with clear records of dataset versions, preprocessing steps, algorithm settings, and evaluation methods.

3)Fairness and Bias Awareness

The system should not favor one language over others and should check if its performance changes for different language groups.

4)Transparency

The system should use models and metrics that are easy to understand whenever possible, and avoid using complex methods without a clear reason.

5)Security of Data Handling

All data processing will happen locally, and raw biometric data will not be shared or made available outside the project.

6)Scalability (Analytical)

The system should be able to test different algorithms and language groups without needing major changes.

7)Academic Integrity

All tools, datasets, and libraries will be properly cited, and results will be reported honestly without overstating security claims.

Design

5.1 Design Architecture

This project uses an evaluation-driven analytical framework instead of a traditional system development lifecycle focused on deployment. The main goal is to build a structured experimental environment for thoroughly testing multilingual keystroke dynamics authentication algorithms, not to create a user-facing authentication product.
Šilingas and Butleris (2015) point out that requirements analysis and design can overlap when using formal modelling methods like UML use-case notation. In this project, this overlap is recognized, but UML use-case modelling is not used because there is no direct human-system interaction to represent. The design instead focuses on data flow, algorithm processing, and evaluation steps, which are better shown through conceptual and pipeline-based design.
The system design is organized around these main components:

- Dataset ingestion and preprocessing
- Feature extraction from keystroke timing data
- Machine learning model training and testing
- Multilingual performance evaluation
- Result comparison and analysis

This modular design lets each component be examined and changed on its own, which helps with reproducibility and controlled experiments.
The overall system architecture follows a linear but modular pipeline design, where data flows sequentially through defined processing stages. Each stage produces outputs that serve as inputs for the next, ensuring clarity and traceability of results.
The main parts of the architecture are:

1. Input Layer: Uses a publicly available keystroke dynamics dataset.
2. Preprocessing Layer: Handles cleaning, normalizing, and sorting data by language.

3. Feature Extraction Layer – Extraction of timing-based keystroke features
4. Model Layer: Applies machine learning algorithms for authentication.
5. Evaluation Layer: Calculates biometric performance metrics.
6. Analysis Layer: Compares results across languages and algorithms.

5.2 Prototype

5.2.1 First Prototype

The main goal of the first prototype is to set up the evaluation pipeline, not to improve performance yet. It is designed to check that the dataset can be processed and that keystroke dynamics features can be reliably extracted in different languages.
This prototype includes:

- Dataset loading and parsing
- Basic preprocessing, such as handling missing values and outliers
- Extracting key keystroke timing features
- Training of a baseline machine learning model
- Running an initial evaluation with standard accuracy metrics

This first prototype acts as a proof of concept, showing that multilingual keystroke data can be managed consistently in one framework.

5.2.2 Second Prototype

The second prototype builds on the first by adding **comparative evaluation**. At this point, several machine learning algorithms are used to compare how well they perform at authentication.
The main improvements are:

- Adding more classifiers
- Separating training and testing sets by language
- Including biometric-specific metrics like FAR, FRR, and EER
- Organizing result logs for later analysis

This prototype makes it possible to compare algorithms in a meaningful way and shows if their performance changes with different languages.

5.2.3 Third Prototype

The final prototype turns the system into a strong evaluation tool. At this stage, the main priorities are rigour, fairness, and reproducibility, rather than just functiona**lity**.
Key improvements include:

- Using the same evaluation protocols for all experiments
- Applying parameter tuning strategies in a consistent way

- Automating how results are gathered and compared
- Preparing outputs that are ready for academic reporting

5.3 Review of technology

5.3.1 Dataset Technology

An existing keystroke dynamics dataset was obtained from a public GitHub repository. The selection criteria for this dataset included the following factors:

- The dataset provides multilingual typing data.
- It meets ethical standards and is distributed under an open license.
- The dataset is of sufficient size to support robust machine learning evaluation.
- Comprehensive documentation of features and structure is provided.

Utilizing an existing dataset mitigates ethical and legal risks related to the collection of behavioral biometric data and ensures compliance with data protection regulations.

5.3.2 Programming Environment

Python was chosen as the primary development language because of its robust ecosystem supporting data science and machine learning applications. The language offers the following advantages:

- Comprehensive and mature libraries for numerical computation
- Extensive machine learning frameworks
- Extensive community support and thorough documentation
- Facilitation of reproducibility through the use of virtual environments

5.3.3 Machine Learning Libraries

Several machine learning libraries were evaluated, including TensorFlow, PyTorch, and scikit-learn. Scikit-learn was chosen as the primary framework for the following reasons:

- Suitability for implementing classical machine learning models frequently applied in keystroke dynamics research
- Comprehensive built-in support for a range of evaluation metrics
- Facilitation of reproducibility and precise parameter control
- Lower computational overhead relative to deep learning frameworks

Deep learning frameworks were considered but excluded at this stage, as the project prioritizes interpretability and comparative evaluation rather than maximizing raw performance.

5.4 Development

Several key development milestones have been accomplished at this stage of the project. A publicly available multilingual keystroke dynamics dataset has been identified and acquired, forming the basis for experimental evaluation. A modular evaluation pipeline has been developed to facilitate systematic preprocessing, feature extraction, model training, and performance analysis. Core preprocessing and keystroke feature extraction routines have been implemented, ensuring consistent data handling across multiple languages. Baseline machine learning models have been trained using these features, and preliminary authentication evaluations have confirmed the feasibility of the proposed approach. Additionally, a structured plan for cross-language performance comparison has been established, supporting controlled and

reproducible subsequent experimentation. Collectively, these achievements validate the evaluation framework and establish a robust foundation for the final experimental phase.

Reflection

This project has offered valuable insights into the technical and conceptual challenges of evaluating behavioural biometric systems, especially in multilingual authentication contexts. A key reflection from this work is the complexity of isolating language-dependent effects in keystroke dynamics while upholding methodological rigor. While keystroke dynamics are frequently described as language-agnostic behavioural traits, both the literature and preliminary evaluations suggest that linguistic structure, keyboard layout familiarity, and cognitive load significantly influence typing behaviour.

From a methodological standpoint, working with publicly available datasets revealed a trade-off between ethical simplicity and experimental limitations. Although using open datasets avoided legal and ethical risks related to data collection, it restricted control over participant demographics, hardware variability, and data balance across languages. This limitation emphasized the necessity of careful preprocessing, normalization, and evaluation design to address confounding factors. The reflection here is that experimental validity in biometric research relies not only on model selection but also on data provenance and feature consistency.

The project also highlighted the limitations of relying exclusively on classification accuracy as a performance metric. Through engagement with the literature, I recognized that biometric systems should be evaluated using metrics such as Equal Error Rate (EER), False Acceptance Rate (FAR), and False Rejection Rate (FRR) to meaningfully assess security and usability trade-offs. This realization influenced the design of the evaluation framework and shifted the focus from algorithmic performance alone to system-level reliability, which is crucial for real-world authentication scenarios.

On a broader level, this work reinforced the ethical responsibilities inherent in biometric system design. Even when direct data collection is avoided, behavioural biometrics present concerns related to surveillance, consent, and potential misuse. Critically engaging with these issues demonstrated that technical effectiveness must be balanced with transparency, proportionality, and fairness. This understanding has shaped a more cautious and responsible approach to system evaluation and design.

In conclusion, this project has contributed technical knowledge about multilingual keystroke dynamics and fostered a deeper understanding of experimental design, ethical awareness, and critical evaluation in biometric research. These reflections will guide both the final stages of this project and future work in security-oriented machine learning research.

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Appendices

Appendix A: Final Project Proposal

Project Title: Enhancing User Authentication through Multilingual Keystroke Dynamics and Machine Learning
Student Name: Orkidea Merko